Final Project

Due: Monday April 20 2026, 11:30pm on AVENUE (there will be no grace period) and **late submission will not be accepted !**

Exercise 1 *Final Project*

As a minimum for a final project you can chose a dataset from `http://archive.ics.uci.edu/ml/index.php` or another repository and model and analyze it using some of the techniques used in the course. You're welcome to go beyond what we have covered in the course and look at techniques such as natural language processing, style transfer in image processing, etc.. As long as it is something that is clearly aligned with neural networks and machine learning it is fine. The final project will count for 25% of the mark for the course. The point is to perform some independent work on a subject that you find interesting. Try to be creative and do something new, work with sound files, images, text, etc.

There are **two main components to the project**: the **code** and the **write up**. The code should be working and should be able to be executed by us, preferably in a jupyter notebook or eventually straight python code. The write up should explain your work and your observations, in particular the insights you have gained from your work. You should also explain the techniques you have used and how they work, for instance you can include code snippets in the write up. The write up should be in pdf format and the page count is limited to 10 pages, less than that is fine. See below.

While it is natural to reuse code in many settings it will not be acceptable to simply submit a python notebook you have found on the internet. You should quote your sources and show your independent usage.

The rules are the following:

1. Group submissions are **not** allowed. An individual project is required. You can ask for help and advice from your friends and from the TA's, but the work you submit should be your own.

2. The code part:

   (a) It is preferable to submit the code part in the format of a jupyter notebook, but straight python code is acceptable. In both cases, you should also submit a pdf version of the code part. This is to make it easier for us to read and understand your code.

   (b) The project has to have a working code that we can execute. So make sure that you upload datasets etc and that figures are reproducible.

   (c) You should not simply copy code from the internet. You can reuse code, but you should show your independent usage and you should quote your sources.

   (d) The code should be well organized and well commented. It should be clear what you are doing and how you are doing it.

   (e) The use of frameworks such as pytorch, tensorflow, keras, etc is allowed and encouraged. You can also use libraries such as scikit-learn, numpy, pandas, etc.

(f) Limited use of generative AI tools such as ChatGPT, Codex, Claude, Gemini, Copilot, etc is allowed for the code part. If you do use them, you need to document it and detail the prompts. By limited use we mean that you should not simply ask for the code to be written for you, but you can ask for help and advice on how to write the code, how to debug it, etc. You can also ask for help on how to use a particular library or framework, but you should not generate the entire project this way.

3. The write up:

   (a) It is not allowed to use generative AI tools such as ChatGPT, Bard, etc to write the write up. The write up should be your own work.

   (b) A significant part of the mark for the project will be based on your explanations and observations in the write up. So make sure your write up explains your work and your insights.

   (c) For the write up, it is important to write an understandable report of your work that can be followed and understood.

   (d) The write up should not exceed 10 pages. If you run over it's OK, we won't throw it out - but try not to.

   (e) The final project will be marked using the following rubric:

4. The rubric:

   (a) Quality and complexity of project 25%.

   (b) Creativity and innovation 25%.

   (c) Quality of coding (correctness is not enough) 20%.

   (d) Quality of writeup 30%.

Disclaimer: Assigning marks according to the above rubric will be based on judgement and comparison between the projects. There is no formula we can provide for achieving full marks.